

Adama Science and Technology University
School of Engineering and Information Technologies
Department of Electrical and Computer Engineering

Exam Date: Apr 2, 2013

Time allotted: 3 hrs

Total Marks: 20%

Instruction

- 1. Read carefully and work out the questions according to their order on the exam sheet.*
- 2. State your assumption if any is used.*
- 3. Clear and neat answer is worthy.*

EEng 3203: Applied Electronics I Mid exam

- 4. Do not use pencil and red pen.*

**PART I: ANSWER THE FOLLOWING QUESTIONS BRIEFLY AND PRECISELY TO
THE POINT**

Q 3 Compare the energy band model of silicon with that of germanium.(2)

Q 1 Briefly explain the difference between intrinsic and extrinsic semiconductor materials.(2)

Q 2 What is the effect of doping on the resistance of the semiconductor? (1)

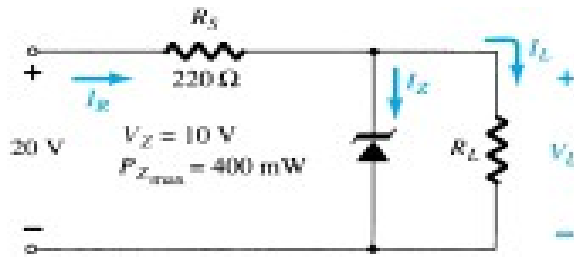
Q4 . Discuss the effect of biasing on the width of the depletion region at a p-n junction.
(2 pts.)

Q5 Discuss the operation of a power supply system with block diagram by showing the
corresponding output waveform from each block. (3 pts.)

**PART II WORK OUT THE FOLLOWING QUESTIONS BY SHOWING YOUR
STEPS CLEARLY.**

Q1.

- (a) Determine V_L , I_L , I_Z , and I_R for the network Fig. below if $R_L = 120\Omega$ (2)
- (b) Repeat part (a) if $R_L = 500\Omega$. (2)
- (c) Determine the value of R_L that will establish maximum power conditions for the Zener diode. (1)
- (d) Determine the minimum value of R_L to ensure that the Zener diode is in the “on” state (1)



Q2. Sketch V_o for the network of Fig. 2 and determine the dc voltage available. (2 pts.)

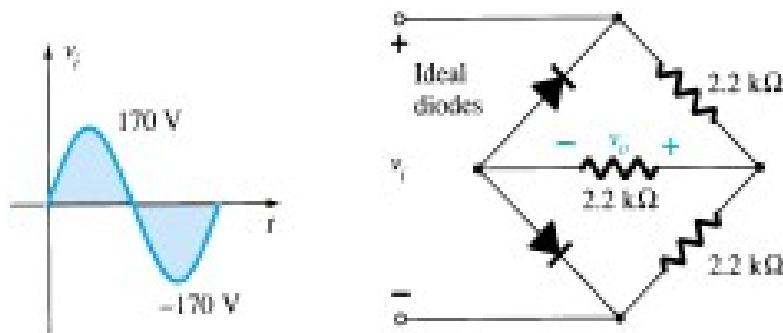


Fig 2

Q3. The secondary voltage has a maximum value of 30 V with 50 Hz and $R_L = 400\Omega$ as shown in fig 3,

- a) What is the dc output voltage & dc current? (2 pt)
- b) Draw the output wave shape characteristics of V_o , V_{D1} , V_{D3} , (1)

- (c) Find the maximum current through each diode during conduction (1)
- d) If a filter capacitor of $C = 220 \mu\text{f}$ is connected in parallel to the load, of $R_L = 400\Omega$, calculate the dc output voltage and dc current? (2pts.)

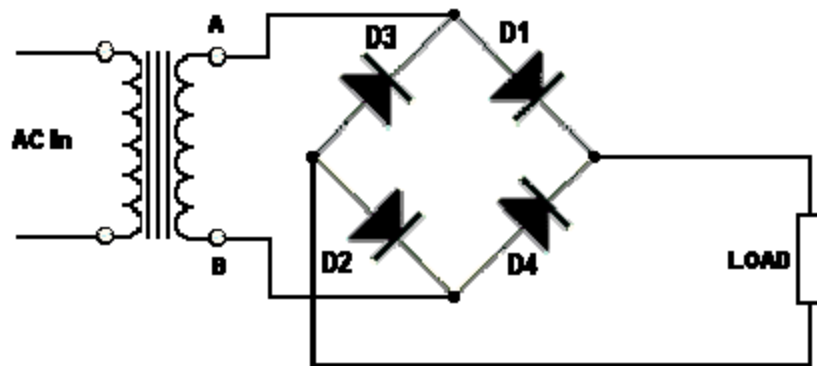


Fig -3

- Q4. Name the circuit and draw the out put waveform for the circuit shown in fig 5 on the space provided with a brief explanation. (assume ideal diode) (1pts.)

